

Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code developed for the **OptiCrop – AI-Based Crop Recommendation System**. The project follows a well-structured coding approach with meaningful variable names, proper file organization, reusable modules, and clear comments. These practices improve code readability, simplify maintenance, and allow future enhancements such as integrating new Machine Learning models, weather forecasting, or additional agricultural features.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation used
2	Proper File Structure	Files and folders are logically organized	Yes	Templates, static files, dataset, model, and scripts are well organized.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables are descriptive and easy to understand.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly represent their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Comments added where needed.
6	Modular Design	Code is split into reusable functions/modules	Yes	Code is divided into modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Duplicate and unnecessary code has been removed.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic validation is implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application	Python(Flask)	Handles routing, user input, and loan prediction	High
2	Machine Learning Notebook	Python,Scikit-learn	Data preprocessing, model training, evaluation, and model saving.	High
3	HTML Templates	HTML/CSS	Home, Prediction, Submit pages	High
4	Trained Model (model.pkl) & Scaler (scaler.pkl)	Joblib / Scikit-learn	Loaded by Flask application for loan prediction.	High
5	CSS Stylesheet	CSS	Used across all web pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for templates, static files, dataset, and notebook.
Readability	5	Meaningful variable names, proper indentation, and easy-to-understand code.
Reusability	5	Reusable Flask routes, trained model, scaler, and HTML templates improve maintainability.
Documentation / Comments	4	Sufficient comments and documentation are provided to understand the application and machine learning workflow.
Overall Score	5	The project follows good coding standards with a modular, readable, and reusable implementation.